

ETHAN LIU

eliu1117-github-io.vercel.app

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Education

University of Maryland, College Park

Bachelor of Science in Computer Science

GPA: 3.73/4.0

College Park, MD

Expected May 2028

Relevant Coursework

Intro to Computer Systems, Object-Oriented Programming II, Algorithms, Programming Language Technologies and Paradigms, Organization of Programming Languages, Introduction to Data Science, Introduction to Machine Learning, Programming Handheld Systems

Technical Skills

Languages: Python, Java, C, TypeScript, JavaScript, Swift, SQL, HTML/CSS, Kotlin, XML, Rust, x86 Assembly

Frameworks/Libraries: Pandas, SciPy, scikit-learn, React, Next.js, SwiftUI, Tailwind CSS, JUnit

Developer Tools: Git/GitHub, Eclipse, VS Code, Jupyter Notebook, Xcode, Android Studio, Replit

Projects

UMD Course Recommender

Next.js, React, TypeScript, Tailwind CSS

- Collaborated in a 4-person team to develop a course recommendation web application for computer science students, integrating REST APIs for live UMD course, section, seat availability, and professor data.
- Engineered a client-side TypeScript parser that transformed degree audit HTML with 20–40 courses per audit into structured JSON containing information about completed courses, in-progress sections, and outstanding requirements.
- Extracted unfulfilled degree requirements from audit data to provide personalized course recommendations from a catalog of 1,000+ general education/computer science courses and a database of 47,000+ course reviews.

Diabetes Risk Prediction from CDC Health Indicators

Python, Pandas, SciPy, scikit-learn

- Cleaned and validated a 70,692-record CDC health survey dataset using Python and Pandas, fixing corrupted features and missing values to allow for reliable use in a 5-member data science project.
- Conducted statistical analysis across 21 health indicators using chi-squared, two-sample t-tests, and Mann-Whitney U tests to identify BMI, blood pressure, cholesterol, and age as significant predictors of diabetes.
- Co-authored the primary technical report, including analysis of logistic regression and SVM machine learning models achieving approximately 74% classification accuracy.

GroupThink

Swift, SwiftUI, Firebase Firestore

- Built an iOS collaborative productivity application in SwiftUI using Apple's Screen Time APIs, applying prompt engineering with Cursor to accelerate debugging and implementation of synchronized application blocking across group study sessions.
- Integrated Firebase for user authentication and synchronization for lobbies and group settings, logging whether users were focused, on break, or distracted during sessions.
- Used Git for version control and Linear for issue tracking while testing, debugging, and iterating on features in a 2-person development team.

Relevant Experience

Casa Chirilagua & InterVarsity USA

June 2026 – July 2026

Web Strategy & Design Volunteer

Alexandria, VA

- Analyzed a professional website audit and researched UX patterns across 15+ nonprofit websites to identify improvements in messaging clarity, donor engagement, and visual accessibility for an organization serving 100+ Latino families.
- Created homepage UI concepts and prototypes informed by peer nonprofit research, working directly with the development team to contribute to a full visual redesign.
- Collaborated with the executive director and development manager to incorporate feedback and align design decisions with the organization's current volunteer recruitment and donor engagement goals.

Organizations

Asian American InterVarsity

August 2026 – Present

Worship Coordinator

College Park, MD

- Lead and organize weekly worship rehearsals and chapter meetings for a 120+ member campus ministry, coordinating logistics and setlists for large-scale events.
- Mentor 20+ worship team members individually to improve musical proficiency and cross-team collaboration through structured practice and feedback.